

Shravya

Role: Sr. Data Engineer

Email: n.shravya.010@gmail.com

Phone: 256-297-4242

SUMMARY

Over 10 years of technical expertise in complete software development life cycle (SDLC), which includes 8 years of Hadoop/Big data/batch & streaming pipelines Development, Administration. Hands-on experience working with Apache Spark with python and Scala and Hadoop ecosystems like MapReduce (MRv1 and YARN), Sqoop, Spark, Hive, Oozie, Flume, Kafka, Zookeeper and NoSQL Databases like HBase.

Apache Spark:

- Excellent knowledge of Spark Core architecture.
- Hands on expertise in writing different RDD (Resilient Distributed Datasets) transformations and actions using Scala, Python and Java.
- Created Data Frames and performed analysis using Spark SQL.
- Knowledge of Spark Streaming and Spark Machine Learning Libraries.
- Worked on File Formats like Sequence files, performed serialization of data on AVRO, parquet file formats and Managing Hadoop log files.

Amazon Kinesis:

- Used for bidirectional near real time sync between Salesforce, Databricks delta lake and Snowflake.
- Implemented AWS Kinesis streams to ingest data into delta lake from the events created on Salesforce.
- Streams are split into micro batches and processed.

Apache Hive:

- Experience in writing queries in HQL, to perform data analysis.
- Created Hive External and Managed Tables.
- Implemented Partitioning and Bucketing in Hive tables for Hive Query Optimization.
- Developed ACID property transactions like Insert, Delete, Update.
- Worked on Conversion of Parquet file into ORC file format to achieve high efficiency while storing and processing.

Apache Oozie:

- Experienced in writing Oozie workflows and coordinator jobs to schedule sequential Hadoop jobs.

Apache Flume and Apache Kafka:

- Used Apache Flume to ingest data from different sources to sinks like HDFS.
- Implemented custom interceptors for flume to filter data and defined channel selectors to multiplex the data into different sinks.
- Excellent knowledge of Kafka Architecture.

- Integrated Flume with Kafka, using Flume both as a producer and consumer (concept of FLAFKA).
- Used Kafka for activity tracking and Log aggregation.

SQL and NoSQL:

- Good understanding of Databases like SSIS, MySQL, Snowflake, AWS RedShift, Postgres.
- Ability to write complex SQL queries to analyze structured data.
- NoSQL database, HBase. Worked on table row key design and to load and retrieve data for real time data processing using HBase API and performance improvements based on data access patterns.
- Knowledge on other NoSQL databases like MongoDB, Cassandra.

Salesforce PRM:

- Data ingestion into salesforce through Bulk Api where each batch max records 10,000.
- Worked and had good knowledge on developer console, Salesforce UI – Object Manager etc.

Apache Sqoop:

- Used Sqoop to Import data from Relational Database (RDBMS) into HDFS and Hive, storing using different formats like Text, Avro, Parquet, Sequence File, ORC File along with compression codecs like Snappy and Gzip.

Version Control and Build Tools:

- Experienced in using GIT, Bit Bucket, SVN.
- Ability to deal with build tools like Apache Maven, SBT.

Java:

- Excellent knowledge of Object-Oriented analysis and design. Very good at analyzing the user requirements and using the design patterns.
- Designed and developed Java enterprise and web applications using Java JDBC API.
- Utilized the concepts of multi-threaded programming in developing applications.
- Implemented unit test cases and documented all the code and applications.

KEY SKILLS

Scala, Python, Java, Linux, AWS, Google cloud, Hadoop – Apache Spark, Spark SQL, MapReduce, Hive, Big Query, Kafka, Amazon Kinesis, Splunk, Flume, Airflow, AFAAS, Oozie, Sqoop, Jenkins, Snowflake, Salesforce

Distributions: Databricks, Cloudera-Cloud break, Horton Works, Amazon EMR, Google DataProc.

PROFESSIONAL EXPERIENCE:

Client: [GE Healthcare, Milwaukee, WI](#)

Role: [Sr. Data Engineer](#)

May 2024 – Current

Responsibilities:

- Designing and Development of Spark-Scala based Data Pipelines, Data Analysis, and Implementation.
- Analyzed existing Spark-Scala pipelines for feature enhancements and bug fixes.
- Utilized AWS S3 cloud Storage for data lake, and stored data files in Parquet and JSON formats as necessary.
- Automated end-to-end ETL workflows using Airflow and scheduled daily batch jobs to ensure reliable data delivery.
- Collaborated with Health Information Exchange (HIE) networks to facilitate secure and efficient data sharing across healthcare systems.
- Ingestion of processed OLAP data into the end application.
- Integrated data tables with AWS Athena for which are used by BI, Tableau and business users.
- Designed and implemented scalable ETL pipelines using PySpark on distributed computing platform as AWS EMR
- Championed the adoption of GitHub Actions for CI/CD, creating streamlined pipelines that integrated testing, code quality checks, and automated deployments across services.
- Used SBT to build Scala Spark project for deployment in aws Code build to run on EMR cluster.
- Worked on version controller GitHub for code base changes.
- Integrated SNS with AWS Lambda and S3 for automated event notifications and data processing in cloud-based ETL workflows.
- Optimized model inference for real-time clinical systems using AWS-based infrastructure, ensuring compliance with HIPAA and data governance standards.
- Integrated and exchanged data across systems using APIs, Snowshare, and flat file formats to ensure seamless data interoperability.
- Worked with Astronomer to deploy and manage Apache Airflow DAGs for orchestrating complex ETL workflows.
- Staged data tables into Postgres (getting migrated to AWS Redshift) and fed into client APIs.
- Followed Agile Methodologies to manage the life cycle of the project. Provided daily updates, sprint review reports, and regular snapshots of project progress.
- Designed fault-tolerant and highly available data pipelines using DynamoDB with AWS Glue, Lambda, and Step Functions.
- Directed cross-functional efforts to build scalable infrastructure and CI/CD pipelines, ensuring reliable software delivery and infrastructure consistency using Terraform and GitHub Actions.

Tools and Technologies: AWS EMR (Spark 3.3), Spark-Scala, Spark-SQL, AWS S3, AWS Athena, AWS Code Build, INTELLIJ IDE, AWS Redshift, Postgres, DynamoDB, GitHub, docker, Jira, Parquet, JSON, CSV, Python.

Client: [T-Mobile, Dallas, TX](#)

Role: [Sr. Data Engineer](#)

Feb 2023 – Mar 2024

Responsibilities:

- Involved in creating Data Pipeline Design, Data Analysis, Develop and Implementation.
- Worked with Hadoop Ecosystem components like HDFS, Big Query, Spark-SQL, on GCP.
- Integrated NLP-based Generative AI systems for transaction classification, fraud detection support, and intelligent customer query handling.
- Develop batch data pipelines using PySpark and Spark SQL.
- Developed shell scripts and Json files for upgrading existing data pipelines and converted them to python based if required.
- Utilized Google cloud Storage based source files for batch ETL processing.
- Ingested processed data into delta tables which are consumed by the downstream data science team.
- Implemented serverless ETL workflows on GCP using Cloud Functions, Dataflow, and BigQuery for real-time and batch processing.
- Integrated identity-aware proxies and encrypted service meshes using Istio for secure inter-service communication in finance applications.
- Developed Airflow DAGs to orchestrate data pipeline jobs.
- Implemented centralized logging and tracing (ELK, Open Tracing) for improved traceability across transaction processing systems.
- Worked on version controller GitHub for code base changes.
- Integrated Looker with Big Query for real-time analytics and self-service reporting.
- Integrated infrastructure-as-code and CI/CD practices across on-prem and cloud hybrid environments to enable faster innovation within legacy telco systems.
- Building Grafana dashboards and populating metrics by consuming data from Big Query tables.

Tools and Technologies: Cloud Break cluster, GCP DataProc, AFAAS, Spark SQL, PySpark, Python, Splunk, GitHub, Apache Kafka, Lenses, Google cloud storage, Big Query, Jenkins, Airflow, Beeline/hive/Big Query, Google cloud.

Client: [Key Bank Cleveland, OH](#)

Role: [Sr. Data Engineer](#)

March 2022 – Dec 2022

Responsibilities:

- Develop batch data pipelines using Databricks, Spark with Scala and Python.
- Utilized AWS S3 based source files for batch ETL processing.
- Ingested processed data into Databricks delta tables, Snowflake and respective Salesforce PRM objects.
- Developed near real time bidirectional sync between Databricks delta lake, Snowflake and Salesforce objects.
- Used Amazon Kinesis to read streams to process data.
- Integrated ETL workflows into CI/CD pipelines using GitHub Actions and Terraform for consistent deployment across environments.
- Developed and maintained ETL pipelines using Fivetran, DBT, and custom SQL scripts to automate data ingestion and transformation processes.
- Integrated Fivetran with DBT to streamline ELT workflows and automate schema changes.
- Leveraged PySpark with Delta Lake for incremental data processing, improving data freshness and reducing storage costs.

- Designed highly scalable pub/sub architectures with SNS and SQS for real-time data streaming in cloud-based analytical applications.
- Developed curl job scripts to pass parameters and run data pipeline jobs.
- Developed Jenkins automated workflows for CI/CD and got notifications to outlook and Slack IM channels.
- Implemented DynamoDB Streams for real-time data processing and seamless integration with AWS Lambda and Kinesis.
- Used MD5 hash for uniqueness and other Databricks related table and job optimizations.
- Worked on version control Bitbucket initially and later moved to Git.
- Ad hoc based S3 files load into staging Snowflake tables.
- Configured SNS topics to trigger SQS queues, Lambda functions, and HTTP endpoints, facilitating scalable and distributed data pipelines.
- Developed ETL pipelines using AWS Glue, Lambda, and Step Functions to process and move data across S3, Redshift, and DynamoDB.
- Developed SQL queries to reflect data on Mobile App which is integrated through MuleSoft.
- Worked closely with BI and Data Science teams to gather requirements on data.
- Installed Docker to Work with Flyway database-migration tool for versioning database schema on Databricks and Snowflake.

Tools and Technologies: Databricks, Spark-Scala, Spark SQL, PySpark, Snowflake, Salesforce PRM, Git, Flyway, Alteryx, AWS S3, Amazon Kinesis, Maven, Jenkins, Docker, Python.

Client: [Vanguard, Malvern, PA](#)

Role: [Big data java developer](#)

May 2021 – Feb 2022

Responsibilities:

- Worked with Hadoop Ecosystem components like HDFS, HBase, Sqoop, Hive, Spark-Scala and Pig with HORTONWORKS Hadoop distribution.
- Utilized Spark SQL to extract and process data by parsing using Datasets or RDDs in Hive Context, with transformations and actions (map, flat Map, filter, reduce, reduceByKey).
- Effectively used GIT (version control) to collaborate with the team members.
- Worked closely with BI and Data Science teams to gather requirements on data.
- Worked on Eclipse with Maven for application development.
- Responsible for the debugging, fixing and testing the existing bugs related to application
- Handled importing of data from various data sources, performed transformations using Hive, Pig, and loaded data into HDFS.
- Implemented DBT testing and documentation, improving data quality and pipeline transparency.
- Optimized SQL models and materializations in DBT to enhance query performance and cost efficiency.
- Performed performance tuning and troubleshooting of MapReduce jobs by analyzing and reviewing Hadoop log files.
- Creating Hive tables to import large data sets from various relational databases using Sqoop and export the analyzed data back for visualization and report generation by the BI team.
- Implemented partitioning, dynamic partitions and buckets in HIVE.
- Participated in Daily Stand-up calls and updated the progress to all the stakeholders.
- Building and deployment of projects using Jenkins.

Tools and Technologies: Hortonworks, Java, Spark SQL and Pyspark, Hive, Sqoop, Bitbucket, Grafana, Jira, DB2, Bash, IntelliJ, Maven, Jenkins.

Client: [Rivian, Plymouth, MI](#)

Role: Big data developer

Nov 2019 – March 2021

Responsibilities:

- Responsible for implementing a generic framework to handle different data collection methodologies from the client primary data sources, validate, transform using Spark (Scala) and load into AWS-S3.
- Designed schema-less data models optimized for key-value and document-based storage patterns in DynamoDB.
- Collects data using Spark Streaming from AWS S3 bucket in near-real-time and performs necessary Transformations and Aggregation on the fly to build the common learner data model and persists the data in HDFS.
- Developed Spark Code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- Migrated historical data to S3 and developed a reliable mechanism for processing the incremental updates.
- Used Airflow workflow engine to manage independent Hadoop jobs and to automate several types of Hadoop such as java MapReduce, Hive and Sqoop as well as system specific jobs.
- Set up different kinds of Build triggers including - Gated -Checking, Continuous Integration. Responsible for Continuous Integration (CI) and Continuous Delivery (CD) process implementation using Jenkins along with Shell scripts to automate routine jobs.
- Coordinate/assist in establishing and applying appropriate branching, labeling/naming conventions using Subversion (SVN) and Code Cloud source control.
- Work with IT marketing analytics to assist with data-related technical issues and provide support
- Created and populated bucketed tables in Hive to allow for faster map side joins and for more efficient jobs and more efficient sampling. Also performed partitioning of data to optimize Hive queries.
- Implemented Curated Data Store logic using Spark Scala and Data frames concepts.
- Explored the usage of Spark for improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark SQL and Spark Yarn.
- Implemented the Spark Scala code for Data Validation in Hive
- Worked on Parquet file format to perform business required jobs.
- Used Spark SQL functions to move data from stage Hive tables to fact and dimension tables.
- Implemented dynamic partitioning in hive tables and used appropriate file format, compression technique to improve the performance of hive jobs.
- Worked with the Data Engineering Platform team to plan and deploy new Hadoop Environments and expand existing Hadoop clusters.

Tools and Technologies: Amazon EMR, Amazon S3, Spark with python and Scala, Hive, Sqoop, Airflow, Design Patterns, SFTP, Code Cloud, Jira, Bash.

Client : [CVS, Irving, TX.](#)

Role: [Hadoop Developer](#)

Feb 2018 – Oct 2019

Responsibilities:

- Hands-on experience in loading data from the UNIX file system to HDFS.
- Transfer of data from landing zone to the HDFS file system using Sqoop.
- Experienced on loading and transforming of large sets of structured and semi structured data from HDFS through Sqoop and placed in HDFS for further processing.
- Designed appropriate partitioning/bucketing schema to allow faster data retrieval during analysis using HIVE.
- Involved in processing the data in the Hive tables using HQL high-performance, low-latency queries.
- Transferred the analyzed data across relational databases from HDFS using Sqoop enabling BI team to visualize analytics.
- Developed custom aggregate functions using Spark SQL and performed interactive querying.
- Managing and scheduling Jobs on a Hadoop cluster using Airflow DAG.
- Involved in creating Hive tables, loading data and running hive queries in those data.
- Extensive working knowledge of partitioned table, UDFs, performance tuning, compression related properties in Hive.
- Work with the Data Engineering Platform team to plan and deploy new Hadoop Environments and expand existing Hadoop clusters.
- Deploy Informatica objects in the production repository.
- Monitor and debug Informatica components in case of failure or performance issues.

Tools and Technologies: CDH5.10, Spark with Python and Scala, Hive, Sqoop, Airflow, Design Patterns, SFTP, Code Cloud, Jira, Bash, UNIX.

Value Labs, Hyderabad, India

Role: Java Developer

Sep 2014 – Oct 2017

Responsibilities:

- Monitor and debug Informatica components in case of failure or performance issues.
- Responsible for analyzing functional specifications and preparing technical design specifications.
- Involved in all Software Development Life Cycle (SDLC) phases of the project from domain knowledge sharing, requirement analysis, system design, implementation and deployment.
- Developed REST web services for implementing the business logic for different functionalities in the features that are developed.
- Utilized CSS, HTML and JavaScript for the development of the front-end screens.
- Wrote Junit test cases for testing the functionality of the developed web services.
- Involved in writing the SQL queries to fetch data from the database.
- Utilized Postman for verifying the smooth workflow of the application, how the application is changing with the newly developed functionalities and verified the output for the web services.
- User login, search & portfolio created using HTML5, CSS3, JavaScript and jQuery.
- Extensively worked on both Enterprise and Community editions of MULE ESB. Experience working with Mule API and Runtime manager and RAML.
- Designed and implemented UI layer using JSP, JavaScript, HTML, DHTML, JSON, XML, XHTML, XSL, XSLT, XSL-FO and business logic using Servlets, JSP, SWING, EJBs and J2EE framework.
- Worked on logging Mechanism Web NMS SNMP API supports logging of the SNMP requests.
- Responsible for the debugging, fixing and testing the existing bugs related to application.

- Developed builds using continuous integration server Jenkins.
- Extensively used GIT for push and pull requests of the code.
- Actively participated in the daily scrum meetings and bi-weekly retro meetings for knowledge sharing.
- Wrote DAO classes using spring and hibernate to interact with databases for persistence.
- Used Eclipse for application development.
- Used JIRA as the task and defect tracking system.